Confined Space Entry Safety

CONFINED SPACE ENTRY SAFETY

The KCP Limited

- A cement plant contains more confined spaces per square meter than almost any other industrial facility.

- Preheater cyclones, raw meal silos, clinker silos, cement silos, ball mills, the kiln shell, cooler compartments, coal hoppers, bag filter housings, process vessels, and underground tunnels — every one capable of killing a worker in minutes.

- Confined space fatalities follow a tragically predictable pattern: insufficient atmospheric testing, expired or missing permits, untrained would-be rescuers who become additional victims. In the cement industry, confined space incidents account for approximately 15% of all workplace fatalities, with a horrifying statistic:

- for every worker who dies in a confined space, 1.5 additional workers die attempting rescue without proper equipment or training. Every single one of these deaths is preventable with proper digital permit enforcement, continuous atmospheric monitoring etc., Raw Meal / Cement Silos: **Risk : Extreme**

Engulfment from material collapse, O_2 depletion from material off-gassing, fall from height (15–40m internal), dust explosion potential. Material can bridge and collapse without warning — standing on a "solid" surface that is actually a thin crust over a void.

- O_2 Depletion
- Engulfment
- Fall (15–40m)
- Dust Explosion

Entry frequency: 2 – 4Times per year (inspection, cleaning, maintenance)

Kiln Shell (during shutdown)

Risk : Extreme

Residual heat (>60°C for 24–48 hrs after shutdown), refractory brick fall hazard, CO/CO_2 accumulation, restricted access through kiln hood or cooler end. Kiln rotation must be absolutely prevented — accidental rotation while workers are inside is immediately fatal.

- CO/CO_2
- Extreme Heat
- Falling Brick
- Rotation Risk

Entry frequency: 1–3 Times per year (refractory relining, inspection)

Preheater Cyclones & Risers

Risk : High

CO gas accumulation (combustion gases persist after kiln shutdown), extreme heat in upper stages, material fall from cyclone walls and buildup, height exposure (60–120m). Access often requires rope work or confined space within confined space (manhole into cyclone from platform).

- CO Accumulation
- High Heat
- Material Fall
- Height (120m)

Entry frequency: 2– 6 Times per year (buildup removal, inspection)

Ball Mills (Raw / Cement) **Risk : High**

Rotation hazard (mill must be locked, de-energized, and mechanically blocked), noise exposure during entry even when stopped (metal structure), respirable dust, restricted egress through small manhole openings. Mill internals (liners, grinding media) present crush and impact hazards.

- Rotation Risk
- Restricted Egress
- Dust Exposure
- Crush/Impact

Entry frequency: 4 – 8 Times per year (liner change, media charge, inspection)

Bag Filters / ESP Housings:

Fine particulate dust (respirable fraction), limited visibility, restricted access and egress, potential energy stored in rapper mechanisms. Accumulated dust on surfaces can become mobile (engulfment risk) if disturbed during cleaning operations.

- Respirable Dust
- Restricted Egress
- Stored Energy

Entry frequency: 4–12 per year (bag replacement, cleaning, inspection)

Atmospheric Monitoring: The Life-or-Death Parameters

Atmospheric hazards kill faster than any other confined space danger — oxygen depletion can cause unconsciousness in 15 seconds and death in minutes. Continuous monitoring isn't optional; it's the difference between a safe entry and a body recovery.

Oxygen (O_2):

Most common confined space killer in cement. Material off-gassing and chemical reactions consume O_2.The Range is 19.5 to 23.5 is Safe to enter into Confined Space.

Below 19.5%: DO NOT ENTER. Ventilate and re-test. If below 16%, treat as IDLH — evacuate immediately if already inside.

Carbon Monoxide (CO):

Accumulates in preheater, kiln, and any space connected to combustion gas paths. Odorless and lethal.

Above 25 ppm: Enhanced ventilation required. Above 50 ppm: Exit immediately, increase ventilation, re-test before re-entry.

- SAFE < 25 ppm
- CAUTION 25–50 ppm
- DANGER > 50 ppm → EVACUATE

Carbon Dioxide (CO_2):

Heavier than air — pools in pits, tunnels, and lower levels of vessels. Cement raw materials release CO_2 during storage.

above 0.5%: Investigate source, increase ventilation. Above 1.5%: Exit space. At 3%+: headache, dizziness. At 4%+: IDLH — rapid incapacitation.

- SAFE < 0.5%
- CAUTION 0.5 –1.5%
- DANGER > 1.5% → EVACUATE

Lower Explosive Limit (LEL) :

Coal dust, fuel oil vapors, and alternative fuel residues create explosion risk in mill, burner, and fuel system spaces.

Above 10% LEL: No hot work, eliminate ignition sources, increase ventilation. Above 25%: Evacuate immediately — explosion imminent at 100% LEL

- SAFE < 10% LEL
- CAUTION 10 – 25%
- DANGER > 25% LEL → EVACUATE

Clinker Cooler Compartments : **Risk : Moderate**

Residual heat from clinker (can exceed 100°C for hours after shutdown), mechanical hazards from grate drive mechanisms, dust exposure, restricted space between grate plates and housing. Hot clinker "red rivers" can persist in dead zones.

- Residual Heat
- Grate Drives
- Dust

Entry frequency: 2–4 Times per year (grate plate replacement, inspection)

Critical rule: Pre-entry testing alone is NOT sufficient. Atmospheric conditions change during work — ventilation can fail, disturbed material releases gases, welding consumes oxygen. Continuous 4-gas monitoring for the entire duration of entry is mandatory. Digital CMMS systems can auto-expire permits if gas monitor readings breach thresholds.

- Before any inspection or entry inside a tank, or air bottle the tank should be thoroughly drained, washed and ventilated. Ventilation should not be confined to purging the atmosphere before entry but in most cases, should be continued throughout the operation.

- All inlet and outlet lines should be isolated. The only satisfactory method of isolation is to disconnect them or to use bolted blanks. Do not rely on valves. The isolation should be supplemented by a notice “Danger Men working in tank”.

- Prior to entry into an enclose space having an atmosphere that may contain toxic or inflammable contaminants or that may be deficient in Oxygen, the atmosphere should be tested with instruments.

- Fresh air mask is to be used for work in atmospheres where the hazard is immediate (toxic flammable or Oxygen deficient). Fresh air masks with more than 150 ft hose or where the inhalation resistance exceeds 2.5 inches of water or the exhalation resistance exceeds one inch of water, should never be used.

- Men working in or making inspection of the interior of the tanks with contained acid and alkali, should be provided with rubber gloves rubber boots, rubber trousers, rubber jackets and acid hoods in addition to respiratory protection. The man entering into the tank should be supplied with a safety belt and lifeline with the free end in charge of an attendant.

- Before starting any work inside acid or oil tank purge it in steam and water.

- Open all manholes and allow the tank for the fresh air entry thoroughly.

- If light is needed, use only flashlights or electric lanterns approved for combustible atmosphere. Use only 24 Volts lamps.

- Wash sludge, sediment and scale from the tank through the drain or remove it with a pump. Flush out well and overflow the tank with water.

- Ventilate or steam the tank for the number of hours required by its size. If steam is used, cool and ventilate (tank up to 48" in diameter requires 12 hrs or more for steaming).

- No entry should be made into the tank if the Oxygen content of the tank is less than 16% Make a gas test and if the tank is found vapor free, check conditions inside the tank before issuing work permit.

- Continue ventilation for duration of the work in the tank, and make periodic tests for the presence of gases or vaporized as the work inside the tank is in progress.

- Where gas or vapor that is likely to be ignited by a spark is present, use tools and equipments of spark resistant material such as wood, fibre, bronze and stainless steel.

- While personnel are entering confined spaces which may be immediately hazardous, a person should always be assigned with no duties other than to observe the personnel in the hazardous atmosphere, tend the lifeline and rescue them if necessary. This person also should be equipped with a harness and lifeline in addition to necessary respiratory equipment.

- Communication should be maintained continuously with personnel inside the tanks.

- Emergency equipment in good operating condition and must be readily available

Thank you